DEEPANSHU SINGH

Sr. R&D Engineer - Applied AI & GenAI

+91 7895285472 **2019deepanshu@gmail.com** **LinkedIn** **Github**

EXPERIENCE

Sr. R&D Engineer | Research and Development
Intervue.io

07/2024 - Present Remote

- **AI Workflow Engineering & Evaluation:** Researched, designed, tested, and implemented **AI/LLM-powered workflows** for **Plagiarism Detection, Coding Evaluation, JD/Resume Parsing, and technical assessment generation**. Designed evaluation criteria and prompts, iterated on outputs, and validated solutions for accuracy, consistency, and contextual relevance.
- **Applied AI & Retrieval Systems:** Developed a **RAG**-based Rubric Intelligence System two-stage retrieval pipeline using **Python**, **Qdrant**, and **BGE-large embeddings**, coupling **Hybrid Semantic Search** with a **Cross-Encoder reranker** and schema validation with **Pydantic** to accurately map unstructured hiring requirements across **10K+** rubric records.
- **LLM Evaluation & Prompt Engineering:** Built systematic evaluation workflows for LLM-powered hiring applications by testing prompts against hundreds of test inputs **across multiple LLMs (GPT, Gemini, Groq, etc.)**. Compared outputs against defined evaluation criteria, analyzed failure cases, and optimized prompts through repeated evaluation cycles to improve reliability and effectiveness.
- **AI-Enabled Automation:** Built workflow automation using **JavaScript** and **Google Apps Script** to streamline assessment creation, validation, and quality-control processes, reducing repetitive manual work and enabling scalable evaluation operations.
- **Research & Evaluation Frameworks:** Defined SOPs for technical research, evaluation-framework design, and data curation, standardizing the translation of evolving technical requirements into structural evaluation criteria across 100+ profiles and 8K+ rubrics.
- **R&D Leadership & Product Integration: Led a 4-member R&D team** and collaborated with Product, Engineering, Customer Success, Product Support, and Operations teams to conduct UAT, analyze user feedback and performance metrics, validate AI solutions, and drive PoCs toward production readiness.
- **Tech Stack:** Python, Prompt Engineering, LLMs, RAG, Embeddings, Qdrant, Hybrid Semantic Search, Vector DBs, Pydantic, JavaScript

R&D Intern
Intervue.io

11/2023 - 07/2024 Remote

- Researched for various roles, skills, tools, and frameworks to develop scalable and structured Rubric-Based evaluation frameworks
- Automated research workflows using JavaScript and Google Apps Script, improving operational efficiency

SDE Intern
ZeMoSo Technologies

02/2023 - 07/2023 Hyderabad

- Developed responsive frontend modules in ReactJS/TypeScript and Material UI, integrating REST APIs for internal workflows.
- **Tech Stack:** HTML/CSS, JavaScript, TypeScript, ReactJS, Material UI, Git

EDUCATION

B.Tech in Electronics & Communication

GL Bajaj Institute of Technology & Management | CGPA **8.2** / 10.0

2019 - 2023 Greater Noida

SKILLS

AI/LLM:

LLMs, Prompt Engineering, RAG, Semantic Search, Hybrid Search, Embeddings, Cross-Encoder Reranking, Qdrant, LLM APIs, **Multimodal AI, Speech-to-Text (Whisper)**, Text-to-Text

Frameworks & Languages:

Python, Asyncio, JavaScript, TypeScript, FastAPI, Streamlit

Automation & Tools:

Workflow Automation, Google Apps Script, Git/GitHub

PROJECTS

Autonomous Finance Tracker AI Agent

[Live Demo] | **[Github]**

- **AI Agent Architecture:** Architected a **multimodal WhatsApp-native AI agent** with both **Speech-to-Text (STT)** and **Text-to-Text (TTT)** capabilities, integrating *whisper-large-v3-turbo* for real-time voice note transcription and chaining all inputs into an LLM parser with strict JSON Schema constraints.
- **Backend & Automation:** Engineered a Supabase (PostgreSQL) backend featuring Role-Based Access Control (RBAC) for in-chat admin approvals, and integrated APScheduler to push automated weekly WhatsApp reports.
- **System Analytics UI:** Built a real-time Streamlit administrative dashboard to visualize live system metrics, track spending trends, and manage user onboarding queues.
- **Tech Stack:** Python, FastAPI, Pydantic, Node.js (whatsapp-web.js), Supabase (PostgreSQL), Streamlit, LLM API, Prompt Engineering, Multimodal AI